LIMO - Silicon Minds | M3 Milestone Report

CS3520 Computer Organisation and Architecture | AY2026/2027

Milestone: M3 - Pipeline Core | Tag: m3 | Release: https://github.com/mohlalanyanepule/LIMO-TEAM-Silicon-Minds-/releases/ta

Live: https://mohlalanyanepule.github.io/LIMO-TEAM-Silicon-Minds-/

1) Objectives Achieved

- Five-stage pipeline (IF,ID,EX,MEM,WB) with pipeline registers
- Control unit and ALU integrated
- Step/Run/Pause/Reset controls
- Register file and data memory views
- Branches resolve in EX with flush of younger instructions
- RAW/load-use hazard stalls
- Self-branch halt
- Datapath UI, control signals, stats
- 112 automated tests passing
- Deployed to GitHub Pages

2) Testing

- npm test: 112 passed
- npm run build: success
- Examples 1-3: produce expected results (x10=48, x10=75, x10=60)

3) Next Steps (M4)

- Forwarding unit (EX/MEM->EX, MEM/WB->EX)
- Load-use with forwarding consideration
- Branch flush visualisation
- Pipeline chart and statistics
- Hazard detection UI